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Academic Trimester: Spring 2026

Session: Summer 2022

Batch/Section: 28th_A(day)

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Department name: Computer Science & Engineering

Faculty Name: Muhammad Anwarul Azim

University name: University of Chittagong.

Course No.: CSE 423

Course Title: Simulation & Modeling.

Course Credits: 3.00

Assignment No.: 05

Online Google Form Submission Date: 05 May,2026

Total Count of Students in this course/class/batch: 61

Total Count of Exercises (or questions or problems): 197

Chapter No. : 12

Chapter Name: Comparison and Evaluation of Alternative System Designs

The problem (question) serial number: 178

The problem (question) Description: Define statistically significant difference and physically significant difference.

Answer: (178) In simulation modeling and analysis distinguishing between statically significant and physically significant is crucial for determining if results are due to change or if they are actually matters.

Statically Significant Difference: A measured difference that is highly unlikely is to have occurred by chance, indicating a real effect within the data (often p value < 0.05). This measures the certainty of the result. It tells us that the difference observed between scenario A and scenario B is not merely "noise" or random variation caused by the simulation's probabilistic inputs. Key drivers: Sample size.

Physically Significant Difference: This measures the importance of the result. It is a difference between simulation results that is meaningful, actionable, or substantial in the real-world system being modeled, clinical or engineering significance. Key drivers: Domain Expertise, effect size.